# Open-Weight Instruction-Tuned and Edge-Capable LLMs for Greek NLP: A Reproducible Evaluation

Anonymous Author(s)

## ABSTRACT

Natural Language Processing (NLP) for lesser-resourced languages faces persistent challenges, including fragmented datasets, inherited biases from high-resource settings, and limited reproducible evaluation protocols. This study addresses these gaps for Modern Greek through three contributions. First, we distill publicly available Greek datasets from prior survey work and classify them according to their suitability for benchmarking, pre-training, and retrieval-augmented generation. Second, we evaluate open instruction-tuned large language models (LLMs) on seven Greek NLP task families for which data are available and legally reusable, revealing task-specific strengths and weaknesses across classification, sequence labeling, translation, correction, and summarization. Third, we provide a reproducible benchmark pipeline that connects dataset selection, model execution, metric reporting, publication-ready artifacts, and automated discovery of candidate datasets from newer literature. Together, these contributions support more transparent evaluation and reuse of Greek NLP resources.

## CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; *Language resources*; • **Information systems** → Data analytics.

## KEYWORDS

Greek NLP, benchmarking, large language models, reproducibility, evaluation

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## 1 INTRODUCTION

Natural Language Processing (NLP) has advanced substantially with the rise of large language models (LLMs), whose development depends on immense volumes of digital data [5]. Although multilingual NLP has benefited from these advances, progress for lesser-resourced languages continues to lag behind that of well-supported languages. As a result, NLP for many of the world's languages often relies on methods, resources, and evaluation practices developed for high-resource settings, inheriting assumptions, biases, and other characteristics that may not align with the properties of less supported languages [**?** ].

This gap is compounded by the scarcity of open-access, high-quality language resources. In this context, high-quality resources refer to native, human-authored corpora, as opposed to synthetic or machine-translated data. Although synthetic and translated resources are often used to bridge data gaps, they can introduce "translationese", where the text inherits syntactic and stylistic constraints from the source language [17], and may lack the lexical diversity and morphological nuance of original language use. Such resources are needed both for pre-training foundation models and for building gold-standard benchmarks for downstream tasks. Comprehensive surveys of NLP resources for lesser-resourced languages therefore do more than document digital readiness: they can also guide future research, reveal reusable datasets, and support the construction of new benchmarks. Our work focuses on one such language, Modern Greek.[1]

In this work, we ask two related research questions: *which properly licensed Greek datasets can be used to train or evaluate LLMs on downstream NLP tasks, and how do open, instruction-tuned, edge-capable LLMs perform across the resulting Greek benchmark?* By addressing these questions, our work makes three contributions:

- Based on a recent Greek NLP survey [**?** ], we extract publicly available Greek datasets that are (a) open-access and (b) permit derivatives, aggregate them into a unified collection that is easy to recompile and use, and provide an automated discovery component for identifying candidate datasets from newer Greek NLP literature.[2]
- We classify the collected datasets according to their usability for benchmarking or pre-training.
- We benchmark open LLMs with manageable parameter counts and Greek-language capabilities across multiple NLP tasks, revealing task-specific strengths and weaknesses.

## 2 RELATED WORK

### 2.1 Greek NLP Resources and Models

Modern Greek NLP has produced a sustained body of task-specific resources and systems over the past decades, including work on grammar checking, legal text processing, named entity recognition, sentiment analysis, and syntactic annotation [1, 3, 11, 26, 31]. More recent surveys have shown, however, that these resources remain unevenly distributed across tasks and domains, and that availability, licensing, and reusability are recurring barriers for reproducible experimentation [24**?** ]. This motivates our focus on datasets that are not only relevant to downstream Greek NLP, but also accessible and legally reusable.

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[1] In our study, the term "Greek" refers exclusively to Modern Greek (ISO 639-1: el), excluding previous stages of the language's history.
[2] The code is available at: anonymised.url

Transformer-based models have also expanded the practical toolkit for Greek. Greek-specific pretrained models such as GreekBERT and GreekBART improved representation learning and sequence-to-sequence modelling for the language [9, 19], while recent open LLMs for Greek, such as Meltemi, indicate growing interest in language-specialised generative models [38]. In parallel, multilingual and open instruction-tuned LLMs such as Llama, Gemma, Aya, Qwen, and Falcon provide broader coverage and are increasingly used through local serving frameworks [2, 7, 14, 36, 42]. Our work complements these model-development efforts by evaluating manageable open models under a shared Greek benchmark protocol.

## 2.2 Benchmarking and Evaluation

Multi-task benchmarks such as GLUE and SuperGLUE established shared evaluation as a central mechanism for comparing general-purpose language understanding systems [39, 40]. Similar efforts have since appeared for individual languages, domains, and multilingual settings, including legal and lower-resource evaluation suites [6, 37, 41]. For foundation models and LLMs, broad evaluation frameworks such as HELM and the LM Evaluation Harness further emphasize reproducibility, task coverage, and transparent reporting [12, 20].

Despite this progress, Greek LLM evaluation remains fragmented across datasets, task definitions, metrics, and execution environments. Existing Greek studies often target individual tasks or model families, making it difficult to compare open instruction-tuned LLMs across heterogeneous downstream settings. Our benchmark therefore prioritizes a reproducible evaluation pipeline over a new modelling method: it brings together properly licensed Greek datasets, standardizes task execution and reporting, and produces outputs that can be inspected, reproduced, and reused in subsequent work.

# 3 THE DATA: FROM COLLECTION TO DISTILLATION

Based on a systematic literature review covering 142 research NLP studies for Greek from 2012 to 2023 [**?** ], we extracted and analysed information on 94 publicly available Greek datasets. Most of these resources were monolingual (84%), but bilingual and multilingual datasets were also present (i.e., with Greek being one of the languages). The most frequent domains were politics and news (20%) while the most frequent task was sentiment analysis (33%).

## 3.1 Availability, Popularity and Licensing

Figure 1 shows the availability status of the collected datasets. The most common status is that **the datasets are not available**, meaning no information is provided by the authors of the respective study regarding their availability. The second most frequent category concerns datasets reported as downloadable, but **actually not publicly available** due to restrictions, such as missing license or subscription limitations. This limitation contradicts the principles and benefits of open access.

Studies with publicly available datasets, serving as a reference point, tend to attract more citations. This is shown by the higher third and fourth quartile of the green box in Figure 2. Furthermore, studies attracting citations while presenting data with limited [15] or no availability [13] belong to the past. The study of Juola and Stamatatos [15] is such an example, shown in Figure 3 with a blue peak in 2013, concerning a shared task on authorship identification (hence, the citations) whose data lack a license (hence, the limitation).

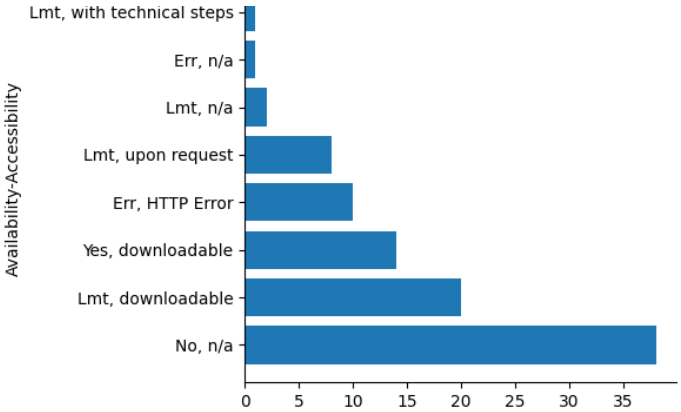


**Figure 1: Availability and accessibility of collected datasets (tuple per bar). Availability is classified to Yes, Lmt (no license or otherwise constrained), Err (hindered by an error), No (not available via a URL). Accessibility is classified to downloadable, HTTP error, upon request, n/a (missing resource); i.e., the outcome we observed when accessing the resource.**

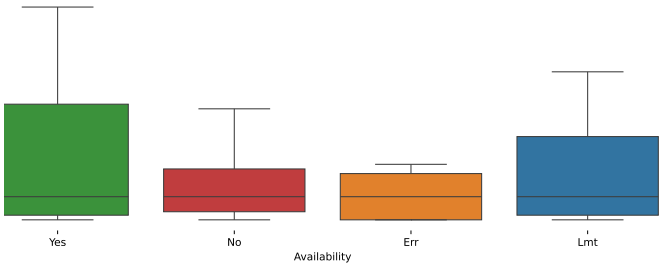


**Figure 2: Whisker and box plots with the number of citations of studies per availability type.**

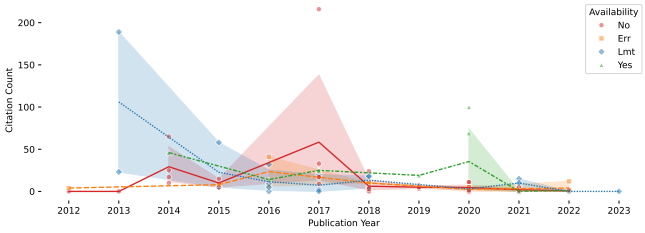


**Figure 3: Citation counts per year of publication of studies developing datasets, based on the availability classified as yes, no, limited (lmt), or erroneous (err).**

## 3.2 AIR Distillation

We collected accessible, interoperable and re-usable (AIR)[3] datasets pooled from a recent survey [? ]. We focused on the 14 Greek datasets that meet our criteria of being licensed, accessible, machine-actionable, and with no hidden costs. We excluded one [10] that was licensed under CC BY-NC-ND 4.0, which does not allow derivatives and thus limits reusability. We also excluded one [43] that shared tweets not via rehydration (i.e., only IDs shared), but as full texts.[4] From the two resources released by Korre et al. [18], we retained the 227-sentence Greek Native Corpus (GNC) for GEC, but excluded the 100-sentence Greek WikiEdits corpus (GWE), which is too small for our study. We also excluded the dataset of open-access e-books [4], because it comprises OCR artifacts.[5] Given these exclusions, we end up with ten datasets, presented in Table 1, which cover a variety of domains.

## 3.3 AIR Resources Suitable for RAG

Table 1 comprises three datasets with no ground truth or whose ground truth is automatically extracted (A/N; i.e., of uncertain quality). Although excluded from our benchmark, these datasets are still possibly useful for retrieval-augmented generation (RAG) and other purposes (e.g., pre-training). These concern 1.28M political speeches, spanning from July 1989 to July 2020 [8]; Greek Wikipedia event pages with named entity recognition (NER) and linking; 101,857 (cleaned) websites with open content [33], including online archives of Greek newspapers from 2003 to 2020, and the Greek part of the W2C corpus [22].

*3.3.1 Dataset statistics.* As shown in Table 2, the text length (measured in characters) greatly varies between the datasets. The dataset with the shortest texts on average is that of Papantoniou et al. [27] and that with the lengthiest is that of Prokopidis and Piperidis [33]. The dataset size (counted in number of files) also varies, starting from ten thousands [27], to one hundred thousands [33], to more than a million [8]. The aggregated text totals up to 1.4 billion characters and 211 million tokens (using white space split).

*3.3.2 Linguistic distance.* We investigate the linguistic associations between these datasets, using statistical language modelling (LMing) as a proxy. For each dataset, we trained one statistical character-level LM on the first 100 characters of 1,000 randomly selected texts per dataset.[6] Then, we computed the Bits per Characters (BPC) across 500 randomly selected texts per dataset, reporting the mean. The result is displayed as a heatmap in Figure 4. High scores (in red) reflecting a greater linguistic distance between the dataset the LM is trained on and the one BPC is computed on. Low scores (in blue) reflect a low linguistic distance, with the lowest scores taking place in the diagonal, when predicting unseen text taken from the same dataset (self-distance). Based on this heatmap, we find that the dataset of Prokopidis and Piperidis [33] shows a higher self-distance compared to the rest while the datasets of Dritsa et al. [8] and Papantoniou et al. [27] are far away from each other, especially when the LM is trained on the former and predicting the latter.

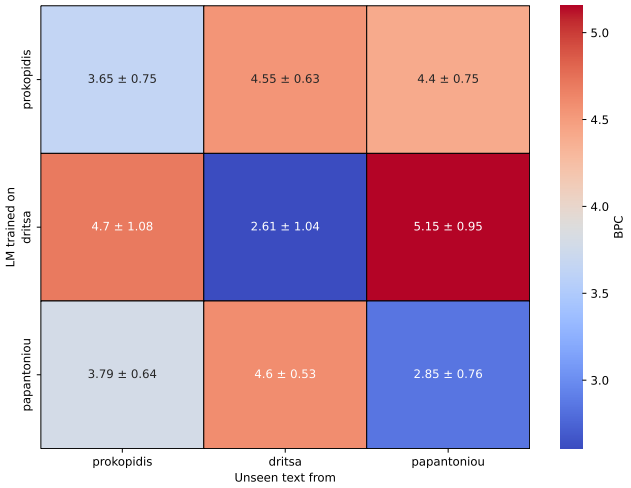


**Figure 4: Heatmap of BPC measured on a sample per RAG dataset (average across the sample; shown horizontally) per trained LM (vertically); greater distance in warm colours.**

## 3.4 Resources Suitable for Downstream Tasks

The datasets used as benchmarks for downstream Greek NLP tasks are summarized in Table 3, along with key statistics and their associated supervised tasks. These resources cover three main categories: classification (Intent, Legal), sequence labeling (POS tagging, NER), and text-to-text generation (Grammatical Error Correction, Machine Translation, Summarization). Next, we present each one.

*3.4.1 Intent Classification.* Student queries, addressed to two university help desks, were manually annotated with six intent labels and three entity types [34]. This dataset is balanced across intent classes, with relatively short texts (average length: 58.6 characters). We use it to evaluate intent classification.

*3.4.2 Legal Text Classification.* A multi-level legal classification dataset is derived from the "Permanent Greek Legislation Code – Raptarchis" [26]. It includes hierarchical annotations of 47 volumes, 347 chapters, and 1,685 subjects. The dataset is highly imbalanced and exhibits large variation in document length (from 17 to approx. 1 million characters). The largest class comprises 34 examples (6.8%; science and arts) and the median has 10 examples. The smallest class has two examples (0.4%; media and tourism). We use it for volume classification.

*3.4.3 POS Tagging.* The Greek Universal Dependencies treebank was part of a project offering standardized treebanks with consistent annotations across languages [25]. This dataset [31] includes

[3] To comply with the FAIR principles, datasets must have a unique and persistent digital identifier (commonly a Persistent Identifier/PID). A PID is a specific combination of numbers and letters that ensures datasets can be uniquely and sustainably accessed as digital objects over time. Merely providing a URL for accessibility does not make a dataset findable, as URLs can become obsolete or break if not properly maintained.

[4] Although it is possible that a company gained access to the texts (e.g., via firehose), releasing them with another license, a user should still be able to delete their content from all platforms. If not, then this dataset is not compliant with the rules. If so, then the dataset is not stable, harming reproducibility of experiments.

[5] Although typos and errors are not prohibited for experiments, we observe that the author names and the title was often injected in the main text, leading to data leaking for potential benchmarks.

[6] Character-level statistical LMing is an efficient way to capture stylistic features, employed in authorship analysis [28].

**Table 1: Greek datasets with their annotation type (AT) defined as manual (M), automatic (A), curated (C; metadata provided by the distributor), hybrid (H; manual and automatic), user-generated (U; from user-edits, not curated) or no (N). We include also the host, size, size unit, domain and license of each dataset.**

| AUTHORS | AT | HOST | SIZE | SIZE UNIT | DOMAIN | LICENSE |
|---|---|---|---|---|---|---|
| Papantoniou et al. [27] | A | Zenodo | 18,615 | DOC | EVENTS | CC BY 3.0 |
| Koniaris et al. [16] | *C* | Hugging Face | 8,395 | DOC | LEGAL | CC BY-NC 4.0 |
| Rizou et al. [34] | *M* | msensis.com | 2,176 | SENT | ADMIN | CC-BY-NC-SA 4.0 |
| Dritsa et al. [8] | A | Zenodo | 1.28M | DOC | POLITIC | CC BY 4.0 |
| Papaloukas et al. [26] | *C* | Hugging Face | 47,563 | DOC | LEGAL | CC BY 4.0 |
| Korre et al. [18] | *M* | GitHub | 227 | SENT | ESSAYS | MIT |
| Bartziokas et al. [3] | *H* | GitHub | 21,153 | SENT | NEWS | CC-BY-NC-SA 4.0 |
| Prokopidis and Piperidis [33] | N | nlp.ilsp.gr | 101,857 | DOC | WEB | CC-BY-NC-SA 4.0 |
| Prokopidis and Papageorgiou [31] | *H* | GitHub | 2,521 | SENT | NEWS, POLITIC | CC BY-NC-SA 3.0 |
| Prokopidis et al. [32] | *U* | nlp.ilsp.gr | 17,018 | DOC PAIR | NEWS | CC BY 4.0 |

**Table 2: AIR datasets for RAG (not in the benchmark) with min, max, mean length in characters and size in documents.**

| | MIN | MAX | AVG | SIZE (#) |
|---|---|---|---|---|
| Papantoniou et al. [27] | 34 | 1,484 | 172.9 | 18,615 |
| Prokopidis and Piperidis [33] | 36 | 67,963 | 2,430 | 101,857 |
| Dritsa et al. [8] | 1 | 291,949 | 900 | 1,280,927 |

syntactic dependencies, POS tags, morphological features, and lemmas. It consists of 2,521 manually validated sentences split into train (1,622), development (403), and test (456) sets.

*3.4.4 NER.* The Greek NER dataset [3] comprises two annotation schemes: a coarse-grained set of 4 entity types (similar to CONLL-2003 [35]) and a fine-grained set of 18 types (similar to OntoNotes [30]). The dataset was developed during a Google Summer of Code project in 2018,[7], where initial automatic annotation was followed by manual curation. The text length varies from 2 to 2,210 characters (Table 3). Both annotation levels exhibit heavy imbalance, with the four-tag level having an imbalance ratio (minor:major) of 0.02 and the 18-tag level showing near-zero imbalance. We assess NER at the less imbalanced four-entity-tag level.

*3.4.5 Grammatical Error Correction (GEC).* The Greek Native Corpus (GNC) [18] comprises (227) student-written sentences, annotated by human experts for (18) error types, related to grammar, spelling, punctuation, and morphology. Each sentence may contain zero, one, or multiple grammatical errors. We use this dataset to assess grammatical error correction.

*3.4.6 Machine Translation (MT).* Bilingual corpora (756 language pairs) exist from content available in Global Voices, where volunteers translate news stories in 41 languages [32]. The 3,629 Greek documents are translated into 40 languages. However, not every document is translated into all languages, resulting in 17,018 bilingual document pairs involving Greek. We use the translations to

[7]https://github.com/eellak/gsoc2018-spacy

**Table 3: Dataset statistics for downstream Greek NLP tasks. We report test set size, number of classes (#C), average text length (in characters), and class imbalance ratio (CIR; smallest-to-largest class proportion) where applicable. Datasets split by us are marked with †.**

| TASK | TEST | UNIT | #CL. | AVG CHAR. | CIR |
|---|---|---|---|---|---|
| Intent Cl. [34] | 436† | sent | 6 | 58.6 | 1.00 |
| Legal Text Cl. [26] | 500 | text | 47 | 3673.3 | 0.07 |
| POS Tagging [31] | 456 | sent | 16 | 138.2 | 0.03 |
| NER [3] | 500† | sent | 4 | 180.9 | 0.02 |
| GEC [18] | 175 | sent | - | 483.0 | - |
| MT [32] | 1,500† | doc | 3 | 125.1 | - |
| Summarization [16] | 300 | text | - | 18,771.3 | - |

English, Japanese, and Farsi, representing high-, moderately-, and low-supported language tiers [**?** ].

*3.4.7 Summarization.* A legal corpus of 8,395 court decisions is compiled from the Greek Supreme Court (Areios Pagos),[8] including the full texts, their summaries, case categories, and associated tags [16]. This dataset exhibits substantial length variability, with 18,541 characters on average and 18,351 st. dev. (min: 2,571; max: 303,538). We use it to assess summarization.

# 4 BENCHMARK SETUP

## 4.1 Tasks and Models

Our benchmark covers seven task families: GEC, intent classification, legal text classification, machine translation, NER, POS tagging, and summarization.

We evaluate five open-weight, instruction-tuned models in the 7B–9B parameter range that are available for local inference through Ollama:

[8]https://www.areiospagos.gr/

- `qwen2.5:7b-instruct` [42];[9]
- `llama3.1:8b-instruct` [14];[10]
- `aya-expanse:8b` [7];[11]
- `gemma2:9b` [36];[12]
- `falcon3:7b-instruct` [2];[13]

This set controls for model scale while maximizing architectural and training diversity. Llama 3.1 and Qwen2.5 provide long-context variants (128K),[14] Falcon3 reports 32K context, and Aya Expanse reports 8K context.

All models are evaluated under the same benchmark pipeline, prompts, dataset splits, and metrics, so observed differences mainly reflect model capability, multilingual transfer, and instruction-following quality rather than evaluation setup.

## 4.2 Deterministic Capped-Test Protocol

The full test sets are very uneven in size, with legal text classification and machine translation being especially large. To keep evaluation tractable while preserving reproducibility, we use a deterministic capped-test profile that always takes the first instances in dataset order (as suggested by the dataset providers) instead of random samples. The resulting capped evaluation (Table 3) contains:

- GEC: 175 examples
- Intent Classification: 436 examples
- Legal Text Classification: 500 examples
- Machine Translation: 500 English pairs, 500 Persian pairs, and 500 Japanese pairs
- NER: 500 examples
- POS Tagging: 456 examples
- Summarization: 300 examples

This design keeps the smaller tasks intact while reducing the runtime burden of the largest tasks. It also supports exact reruns because the selected subset is deterministic. For machine translation, the benchmark now uses a shared data-wrapper path rather than a separate task-local split implementation, which eliminates an earlier source of inconsistency in language-specific counts.

## 4.3 Metrics

We report task-appropriate primary metrics. For GEC, the primary metric is GLEU [23]; we also report exact match and edit-distance diagnostics (WER/CER) against reference and source text. For intent classification, legal text classification, NER, and POS tagging, macro-F1 is the primary comparison metric, with accuracy as a supplementary metric; for NER we additionally report entity-level F1. For machine translation we use chrF [29] as the primary metric and retain WER/CER as supplementary string-distance diagnostics. For summarization we use BERTScore-F1 [44] as the primary semantic comparison metric and report WER/CER and ROUGE [21] as supplementary overlap-based diagnostics. Latency is reported as average per-instance generation time in seconds.

[9] https://huggingface.co/Qwen/Qwen2.5-7B-Instruct
[10] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
[11] https://huggingface.co/CohereLabs/aya-expanse-8b
[12] https://huggingface.co/google/gemma-2-9b
[13] https://huggingface.co/tiiuae/Falcon3-7B-Instruct
[14] An extension of Llama-3.1 is Llama-Krikri-8B-Instruct. Preliminary runs showed mixed task-level behavior, but we omit these results from the main comparison to keep the evaluation focused on distinct model families under the same capped protocol.

To compare models across heterogeneous metrics, we additionally compute a direction-aware normalized quality score per benchmark segment. Within each segment, the best model receives the highest normalized value and the worst model the lowest. These segment-normalized scores are then averaged to obtain an overall ranking. The main paper focuses on the primary metrics used for ranking, while the appendix reports additional prompt and winner-margin material.

# 5 EXPERIMENTAL RESULTS

We analyse the experimental results by focusing on the best model per task, but also touching on the relationship between performance and efficiency. Table 4 summarizes the best model in each benchmark segment. Figure 5 visualizes normalized quality across segments, while Figure 6 shows the trade-off between quality and inference speed. Furthermore, we provide an analytical table per task.

## 5.1 Patterns across models

*5.1.1 The best model.* The strongest overall model is `gemma2:9b`, ranking first on the normalized aggregate with an average quality score of 84.4 and winning intent classification, legal text classification, NER, POS tagging, and two of the three machine translation segments.

*5.1.2 The runner up complements the best.* `llama3.1:8b-instruct` ranks second overall with an average normalized quality score of 53.8. It wins GEC and remains highly competitive in intent classification and summarization, but it trails `gemma2:9b` on legal text classification and the aggregate ranking.

*5.1.3 Competitive in cross-lingual tasks.* `qwen2.5:7b-instruct` is third, which is especially competitive on Persian machine translation and wins that segment. `aya-expanse:8b` ranks fourth overall, but remains a strong specialist in summarization and is the runner-up on legal text classification and English translation.

*5.1.4 The weakest model is not uniformly weak.* The weakest model is `falcon3:7b-instruct`. This is often among the fastest models even when it lags in quality, which reinforces the value of reporting latency alongside task scores.

## 5.2 Patterns across tasks

*5.2.1 Text classification.* Legal text classification remains a difficult benchmark component. As shown in Table 4 and the detailed results in Table **??**, the refreshed coarse-grained setup raises the best macro-F1 to 0.126 with `gemma2:9b`.[15] This suggests that the task is still demanding under zero-shot prompting and may benefit from better label descriptions, exemplars, or task-specific fine-tuning.

By contrast, intent classification is comparatively more tractable. As reported in Table 4 and Table **??**, `gemma2:9b` achieves a macro-F1 score of 0.737, substantially outperforming other models on this task.

*5.2.2 Translation.* The capped profile evaluates 500 examples per target language, which is enough to separate models consistently while keeping runtime manageable. As summarized in Table 4 and

[15] We replaced opaque identifiers with coarse-grained volume labels.

**Table 4: Best-performing model per benchmark segment on the deterministic capped evaluation run (GEC=175, Intent=436, Legal=500, MT(ENG=500, FAS=500, JPN=500), NER=500, POS=456, Summarization=300), using representative metrics per task.**

| Task Segment | Winner | Primary Metric | Value |
|---|---|---|---|
| GEC | Llama 3.1 8B Instruct | GLEU | 0.796 |
| Intent | Gemma2 9B | Macro-F1 | 0.737 |
| Legal | Gemma2 9B | Macro-F1 | 0.126 |
| MT-ENG | Gemma2 9B | chrF | 61.781 |
| MT-FAS | Qwen2.5 7B Instruct | chrF | 18.325 |
| MT-JPN | Gemma2 9B | chrF | 16.726 |
| NER | Gemma2 9B | Macro-F1 | 0.122 |
| POS | Gemma2 9B | Macro-F1 | 0.486 |
| Summ. | Aya Expanse 8B | BERTScore-F1 | 0.519 |

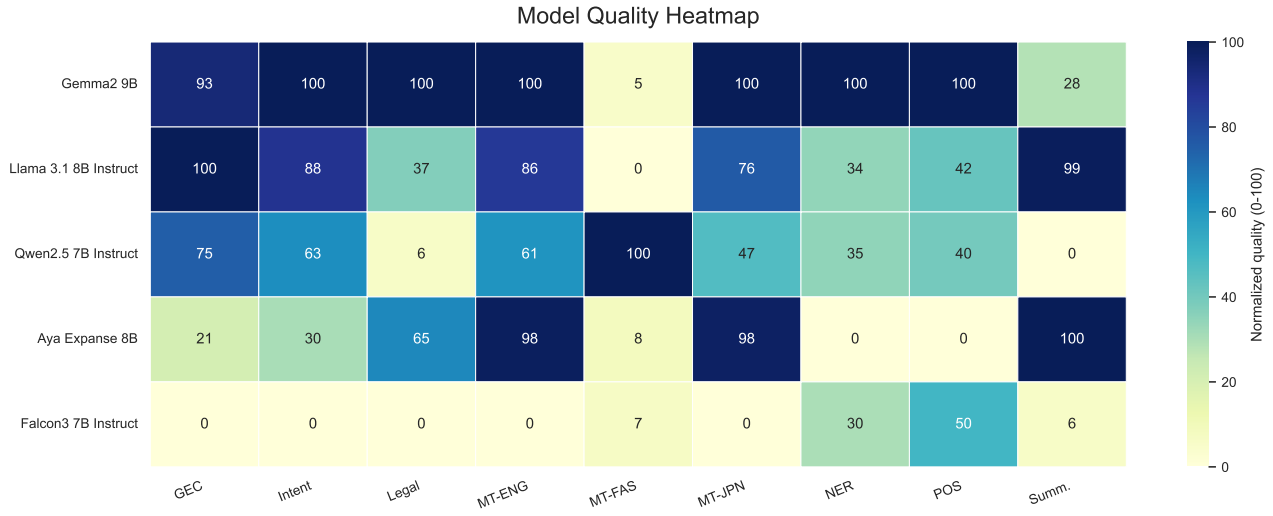


**Figure 5: Normalized quality heatmap across the nine benchmark segments. Scores are normalized within each segment so that higher is always better, even when the underlying metric is lower-is-better such as WER.**

detailed in Table **??**, `gemma2:9b` achieves the highest chrF score for English and Japanese translation. Notably, its chrF score exceeds the rest of the field despite not achieving the best WER, indicating that chrF (character-level) and WER (word-level) capture different aspects of translation quality and can rank models differently.

`qwen2.5:7b-instruct` clearly outperforms other models on Persian translation (Table **??**), indicating that some models (notably Qwen) are better equipped to handle lower-resource or typologically distant language pairs such as Greek–Persian.

*5.2.3 POS tagging and NER.* As shown in Table 4 and the detailed results in Tables **??** and **??**, `gemma2:9b` leads on macro-F1 in both POS tagging and NER. However, `llama3.1:8b-instruct` achieves higher Accuracy in NER (but not in POS tagging), suggesting differences in handling minority entity classes and class imbalance.

*5.2.4 Summarization.* The winner is `aya-expanse:8b`, but only by a very small margin from `llama3.1:8b-instruct` (0.519 vs. 0.517), which is also better in WER. This suggests that the top two models are effectively near-tied semantically.

## 6 DISCUSSION

The benchmark combines multiple Greek NLP settings in a single reproducible workflow, but the current evaluation still has limitations. The capped protocol is intentionally pragmatic rather than exhaustive. Although well suited for iterative comparison, it is not a substitute for full test-set evaluation when final archival numbers are required. Specific metric choices are also provisional. In particular, summarization and machine translation would benefit from stronger semantic metrics and human assessment, while legal text classification would benefit from richer label semantics. Even so, the supplemental metrics improve interpretability relative to a WER-only or macro-F1-only summary.

Despite these caveats, the benchmark exposes meaningful structure in model behaviour. Some models offer strong general performance across multiple tasks, while others are better understood as specialists for translation or summarization. This kind of cross-task perspective is hard to obtain from isolated task runs and is exactly where a unified benchmark becomes valuable.

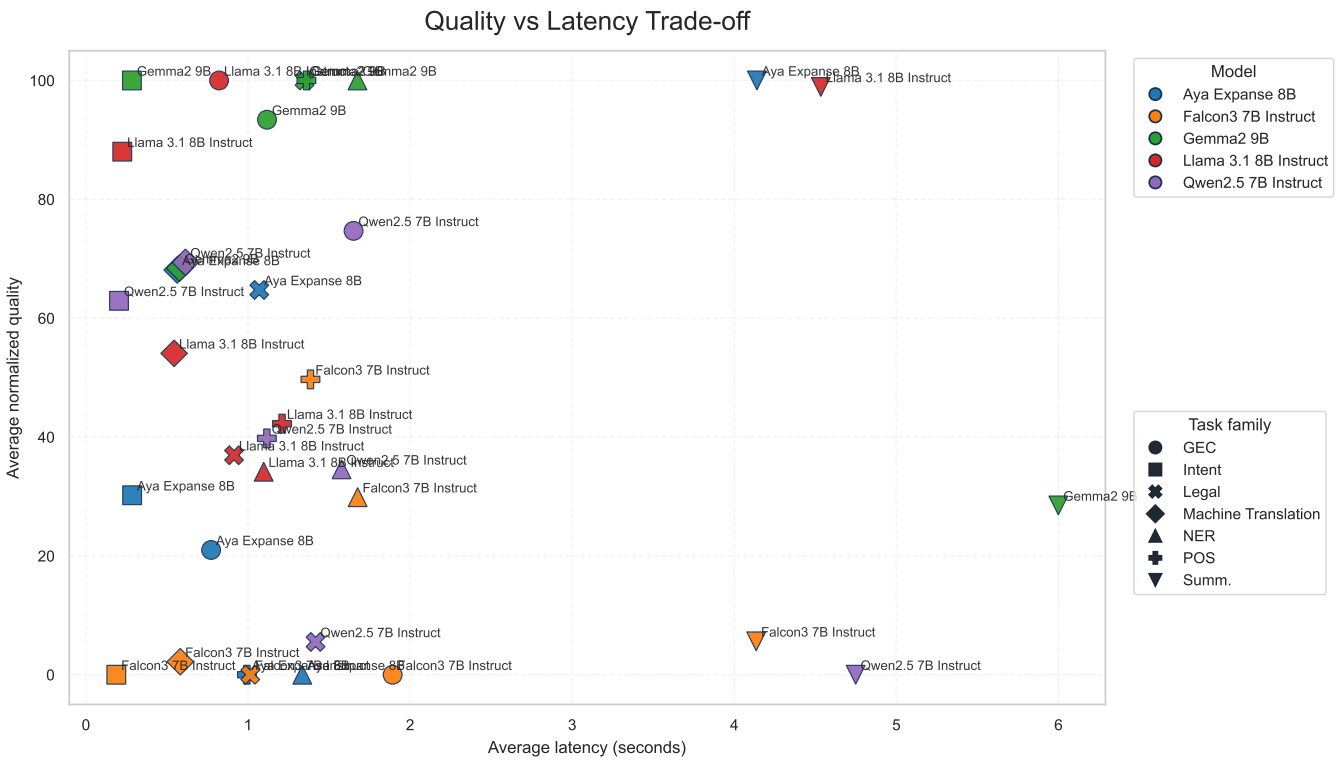


**Figure 6: Quality-latency trade-off across models and task families. Each point corresponds to one model-task pair after averaging within the task family.**

## 7 EVERGREEN DATASET DISCOVERY

Beyond our static benchmark, Greek NLP resources continue to evolve. To make the resource collection maintainable rather than fixed to the 2012–2023 survey, we implement a lightweight evergreen discovery pipeline for newly published work. This component is not used to define the benchmark scores reported above; instead, it demonstrates how the resource inventory can be updated as new datasets appear. To support the continuous expansion of Greek NLP benchmarks, we adapt the search and screening protocol of **?** ] to recent literature (2024–2026) retrieved from the *ACL Anthology* and *Semantic Scholar* ($n$ = 118).[16] The objective is to provide a proof of concept for an *evergreen* pipeline for identifying new Greek NLP datasets.

The screening process was applied in two stages. First, non–peer-reviewed publications were removed using rule-based filtering. The remaining exclusion criteria were then handled with a zero-shot LLM classifier. This process yielded 55 in-scope papers, of which 48 were identified as likely containing dataset references based on indications in the abstract that a dataset was introduced or released.

For these papers, dataset URLs were recovered automatically through a retrieval pipeline: article text was extracted, URLs were identified with regular expressions, and *Llama 3 70B* selected the most plausible dataset link. This produced 18 candidate dataset URLs, 13 of which were verified as reachable.

A manual audit on a small dataset-annotated subset ($n$ = 18) showed high precision and moderate recall (precision = 1.00, recall = 0.69), indicating a conservative but reliable approach. Finally, titles and abstracts were classified into `canonical_task` categories using *Llama 3 70B*, enabling task-level organization of newly identified resources, mainly in text classification and machine translation.

Overall, the results indicate that automatic mining can effectively complement manual curation and support the systematic, continuous enrichment of Greek NLP resources.

## 8 CONCLUSION

We presented a reproducible Greek NLP benchmark for open, instruction-tuned, edge-capable LLMs and used it to compare five models across seven task families and nine benchmark segments. Under a deterministic capped-test protocol, `gemma2:9b` ranks first overall, while `llama3.1:8b-instruct`, `qwen2.5:7b-instruct`, and `aya-expanse:8b` emerge as specialists on GEC, Persian translation, and summarization respectively. The current pipeline is already useful for comparative evaluation, and it also provides a clean path from benchmark execution to publication assets. Future work should extend the benchmark with richer label semantics for legal text classification, stronger semantic metrics and human assessment for translation

[16]The implementation is available in our GitHub repository, enabling reuse and continuous application of the protocol for future dataset discovery.

and summarization, and continuous dataset discovery to keep the Greek NLP resource collection up to date.

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**Table 7: Legal Text Classification.**

| | |
|---|---|
| System | You are a Greek legal text classification system. Return only one label from the provided label set and nothing else. |
| User | Given the following Greek legal text, return only its coarse-grained legal volume label. Choose exactly one label from: <LABEL_SPACE><br>Text: <TEXT><br>Label: |

**Table 8: Machine Translation.**

| | |
|---|---|
| System | You are a translation system. Translate the user's text accurately. Return only the translation and nothing else. |
| User | Given a text in Greek, translate it to <TARGET_LANG>. Only the translation should be returned.<br>Text: <SOURCE_TEXT><br>Translation: |

**Table 9: Named Entity Recognition.**

| | |
|---|---|
| System | You are a Greek named entity recognition system. Return only a Python-style list of tags and nothing else. |
| User | You are given a tokenized Greek sentence. Return only a Python-style list of NER tags, one tag per token. Use only these labels: <NER_LABEL_LIST><br>Tokens: <TOKEN_LIST><br>Tags: |

**Table 10: POS Tagging.**

| | |
|---|---|
| System | You are a Greek POS tagging system. Return only a Python-style list of tags and nothing else. |
| User | You are given a tokenized Greek sentence. Return only a Python-style list of POS tags, one tag per token. Use only these labels: <POS_LABEL_LIST><br>Tokens: <TOKEN_LIST><br>Tags: |

**Table 11: Summarization.**

| | |
|---|---|
| System | You are a Greek summarization system. Produce a concise summary in Greek and return only the summary. |
| User | Given a Greek legal text, provide its summary also in Greek. Only the summary should be returned.<br>Text: <DOCUMENT_TEXT><br>Summary: |


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# A PROMPT TEMPLATES

The following tables list the system and user prompt templates used in the benchmark.

**Table 5: Grammatical Error Correction (GEC)**

| | |
|---|---|
| System | You are a careful grammatical error correction system for Modern Greek. Correct the user's sentence while preserving its meaning. Return only the corrected Greek sentence and nothing else. |
| User | Correct the following Greek text grammatically and orthographically. Preserve the same meaning.<br>Text: <ORIGINAL_TEXT><br>Corrected text: |

**Table 6: Intent Classification.**

| | |
|---|---|
| System | You are a Greek intent classification system. Return only the intent label and nothing else. |
| User | Given a text, provide the intent of the text. Only the intent should be returned.<br>Possible intents: <LABEL_SPACE><br>Text: <TEXT><br>Intent: |